

Measuring the Dark-Matter Scale Height from Stellar Phase Space: A Summary of the Simulation–Recovery Experiments

July 12, 2026

1 The aim of the project

We want to measure the **dark-matter scale height** h_{dm} of the Galactic disk from the vertical phase-space distribution of stars, (z, w) (height above the midplane and vertical velocity), and—crucially—to understand *when and why* that measurement is biased.

The pipeline is a controlled, end-to-end numerical experiment:

1. **Generate** an initial stellar distribution function (DF) $f_0(z, w)$ at time $t = 0$.
2. **Propagate** every star through a gravitational potential $\Phi(z)$ by Hamiltonian (leapfrog) flow to the observation time t_{obs} , producing the “observed” phase-space cloud.
3. **Recover** the potential parameters—chiefly h_{dm} —by *back-integrating* the observed stars through a trial potential and maximising the likelihood of the implied initial conditions. The whole back-integration is differentiable, so the fit is done by gradient descent (PyTorch autograd).

Because we generate the data ourselves, the ground truth is known exactly, so any gap between the recovered and true h_{dm} is a *controlled bias* we can dissect.

The key physical difficulty. The dark-matter contribution to the potential is

$$\Phi_{\text{DM}}(z) = 4\pi G \rho_{\text{DM}} h_{\text{dm}}^2 \log \cosh\left(\frac{z}{h_{\text{dm}}}\right).$$

Expanding near the midplane, the curvature $\Phi''_{\text{DM}}(0) = 4\pi G \rho_{\text{DM}}$ is **independent of** h_{dm} ; the scale height enters only through the *anharmonic* terms ($\sim z^4/h_{\text{dm}}^2$ and higher). Hence h_{dm} is imprinted only on high- z / high-action stars and is intrinsically *weakly constrained*. This single fact explains most of the results below: anything that perturbs the high- z tail (a wrong DF shape, an unmodelled transient) can masquerade as a change in h_{dm} , and a slowly time-varying $h_{\text{dm}}(t)$ is nearly degenerate with a static end-point value.

2 What each test does

Test 1 — Mismatched DF (a deliberate failure). Data are generated with one DF; the fit assumes a *different* DF (e.g. a mismatched velocity dispersion σ). This shows that an incorrect assumption about the stellar distribution biases h_{dm} . A first attempt to “patch” the mismatch with a small time-independent neural-network (NN) correction to the potential is included—and shown to be insufficient.

Test 1b — Bias asymmetry, quantified. A verification of the direction and size of the Test-1 bias using the leading r^2 (harmonic) term and an exact-formula effective-radius fit. It confirms the bias is a real, predictable consequence of the DF mismatch, not a numerical artefact.

Test 2 — Can a time-dependent potential remove the bias? One shared, biased dataset is fit with two different *time-dependent* corrections, swept over an amplitude cap α : (a) a physically-motivated **passing-satellite** model, and (b) a generic **Fourier** $h_{\text{dm}}(t)$. *Lesson:* extra time-dependent freedom removes the bias *only if it has the right functional form* (the satellite), not if it is a generic $h_{\text{dm}}(t)$. A smooth $h_{\text{dm}}(t)$ barely changes the likelihood—the data are nearly blind to it (the anharmonic/adiabatic degeneracy above)—so it cannot absorb a localised transient.

Test 3 — Give the fit an honest, larger freedom. The fit is handed the *full* single-Gaussian DF ($\mu_z, \mu_w, \sigma_z, \sigma_w$ all free) *plus* a small NN potential bump capped at 10% of the static potential peak, on a *clean* dataset (static halo, no satellite, truth known exactly). Question: does this extra freedom still return the correct DF ($\mu \approx 0$, $\sigma \approx \text{truth}$), the correct $h_{\text{dm}} \approx 250$ pc, and a near-zero NN? This tests whether flexibility *preserves* the answer when there is no real feature to absorb.

3 The two new tasks

These extend Test 3 by relaxing the assumption that we know the shape of the stellar DF—the honest stress test of the whole method.

Task 1 — Double-Gaussian DF + time-dependent potential $\Phi(x) + \delta\Phi(x, t)$ (implemented). The initial DF becomes a **two-component (cold + hot) Gaussian mixture** with *every* property free—both means and both dispersions of each component, plus the mixing weight π —and the potential gains a **time-dependent** correction $\delta\Phi(z, t) = A_{\text{nn}} \text{MLP}(z/z_{\text{max}}, t/t_{\text{end}})$, capped at 10% of the static peak. On a clean two-population dataset with a static halo, we ask whether the joint fit still recovers the true two-population DF *and* the correct h_{dm} , with $\delta\Phi$ staying small. This checks that added DF freedom fixes DF-shape bias without opening a new degeneracy with $\delta\Phi(x, t)$.

Task 2 — Normalizing-flow DF + $\Phi(x) + \delta\Phi(x, t)$ (planned). Task 1 pushed to the limit: f_0 becomes a **normalizing flow**, a neural density model able to represent *any* smooth initial distribution. The dynamics are still forced to come entirely from the potential $\Phi(x) + \delta\Phi(x, t)$ —the flow describes only the initial condition. The question “*does it still return the correct h_{dm} ?*” is genuinely non-trivial: a maximally flexible DF could in principle make h_{dm} unidentifiable. If h_{dm} survives, the headline result follows: **the dark-matter scale height can be recovered without assuming the stellar distribution function.**

4 The through-line

Every test circles the same tension:

- Flexibility of the *wrong shape* (a mismatched DF, a generic $h_{\text{dm}}(t)$) **biases** h_{dm} or fails to fix an existing bias.
- Flexibility that is *too generic* risks making h_{dm} **degenerate**.

- The goal is to find the freedom that is “just right”—enough to absorb our ignorance of f_0 , structured enough to leave the dynamical imprint of h_{dm} intact.

Tasks 1 and 2 are the decisive experiments: they test whether letting f_0 be flexible removes DF-shape bias *without* creating a new degeneracy with the time-dependent potential.